6 Nuclear physics

2012-6.

d.

(i) Name two types of radioactivity. (2marks)

(ii) Define "half- life" of a radioactive element. (1mark)

(iii) Mention any three properties of an alpha particle. (3marks)

2012-8.

d. Explain any two uses of nuclear radiation. (4marks)

2011-7.

d.

(i) Define "nuclear fission". (1mark)

(ii) Write down two symbols representing isotopes of hydrogen. (2 marks)

e. Explain why a charged leaf electroscope is not suitable for detecting beta radiation. (2 marks)

f. Thorium $({}^{232}_{90}Th)$ decays by alpha emission to radium $({}^{228}_{88}Ra)$.

(i) What do the numbers 228 and 88 represent in $({}^{228}_{88}Ra)$?

228: (1 mark)

88: (1 mark)

(ii) Write the nuclear equation for the decay of thorium $({}^{232}_{90}Th)$. (3 marks)

2010-5.

a.

(i) Name two particles found in the nucleus of an atom. (2 marks)

(ii) State two characteristics of a radioactive substance. (2 marks)

(iii) State any two uses of nuclear radiation. (2 marks)

b. **Table 1** shows the activity of a radioactive element with time.

Table 1

Activity (Disintegrations per second)	40	30	20	10
Time (days)	0	4	8	16

(i) Plot a graph of activity against time. (5 marks)

(ii) Use the graph to find the time taken for the activity to drop from 30 to 15 disintegrations per second. (2 marks)

c. Name the particles emitted when radium $({}^{226}_{88}Ra)$ decay to radon $({}^{222}_{86}Rn)$. (2 marks)

2009-3.

a. Define "radioactivity". (2marks)

b. Name three types of radiation. (3marks)

c. Mention any two instruments that are used to detect radiation. (2marks)

d.

(i) How do chemical properties of the isotopes of uranium, ${}^{238}_{92}U$ and ${}^{234}_{92}U$ compare? (1mark)

(ii) Give a reason for the answer to 3.d.(i). (2marks)

e. Mention any one natural source of radiation. (1mark)

f. A radioactive source has a half-life of 30 minutes. Calculate the fraction left after 2 hours. (4marks)

2008-3,

a. Define "nuclear fusion". (2 marks)

b. The equation below shows the fusion of heavy hydrogen

$${}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$$

(i) Name the products:

${}^{3}_{2}\mathrm{He}$ (1 mark)

${}^{1}_{0}\mathrm{n}$ (1 mark)

(ii) Identify the type of radioactive decay. (1 mark)

c. Uranium, ${}^{238}_{92}U$, decays in a series of stages as follows:

Stage 1; ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th$

Stage 2; ${}^{234}_{90}Th \rightarrow {}^{234}_{91}Pa$

(i) Which particles are emitted at each stage?

Stage 1; (1 mark)

Stage 2: (1 mark)

(ii) Apart from the particles mentioned in 3.c.(i), what else is emitted at each stage? (1 mark)

d. State any three safety precautions that must be followed when handling radioactive substances. (8 marks)

e. When a source of radiation is placed in front of a Geiger-Muller counter, the initial count rate is 128. After 16 minutes, the count rate is 8. Calculate the half-life of the source. (6 marks)

2007-8.

Describe the following radioactive processes: alpha decay, beta decay and gamma emission. In the description include atomic numbers, penetration power, ionizing ability and behavior in magnetic and electric fields.

Alpha decay: (5marks)

Beta decay: (5marks)

Gamma emission: (4marks)

2006-3.

a. Define "nuclear fission". (2marks)

b. Name two particles found in the nucleus of an atom. (2marks)

c. Two radioactive samples showed the following characteristics.

Sample	Effect on gold leaf electroscope	Effect of card board on sample
A	No effect	Sample passes through
B	Discharged	Sample blocked

(i) Identify particles emitted by samples A and B. (1mark each)

(ii) Why does the particle emitted by sample B get blocked by the card board? (2marks)

d. **Figure 1** is a diagram showing radiation passing through an electric field.

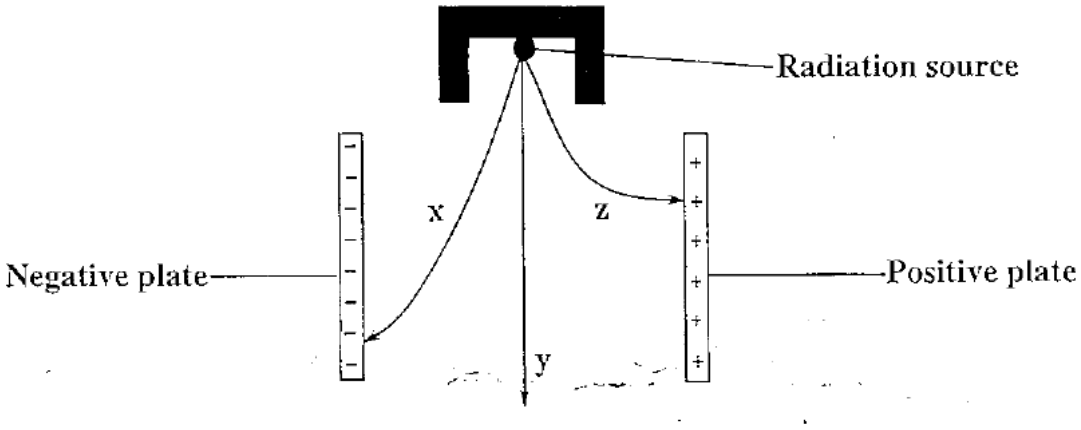

Figure 1

(i) Name the particles taking paths **X**,**Y** and **Z**. (1mark each)

(ii) Explain why particle **Z** will deflect towards the positive plate. (3marks)

2005-4.

c.

(i) the half-life of a radioactive substance is 3 hours. What mass of the substance would remain after 12 hours if the initial mass was 20g? (4marks)

(ii) Why is it important to use radioisotopes with a short half life as tracers in Agriculture? (2marks)

(iii) Explain why fission is a useful process in industry. (3marks)

2004-6.

a. State any two safety precautions to be taken when handling radioactive substances. (2 marks)

b.

(i) Describe the "alpha particles" and "beta particles"

alpha particles

beta particles (6 marks)

(ii) Explain why gamma radiation is used in sterilization of medical equipment. (3 marks)

c.

(i) The Radon-222 (Rn) isotope is formed from the alpha decay of radium-226 (Ra).

Write a nuclear equation to show this change. (Atomic number of Ra is 88). (3 marks)

(ii) **Figure 4** is a diagram showing radiation passing through ax; electric field. Draw and label in the diagram the paths taken by the alpha-particles and the beta-particles. (2marks)

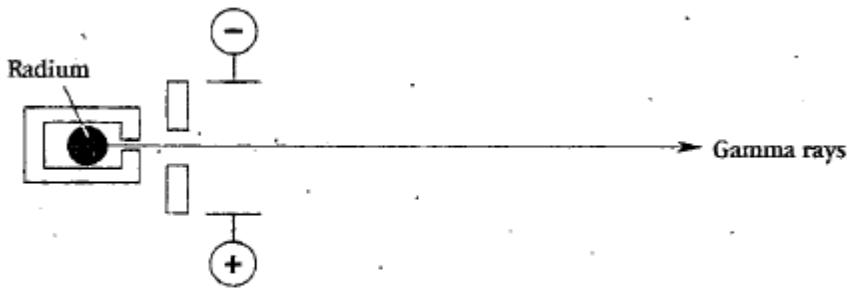

Figure 4

(iii) Explain Why the behavior of particles is as shown in 6c(ii). (4marks)

2003-2.

a What are isotopes? (2marks)

b. Chlorine has two isotopes $^{35}_{17}Cl$ and $^{37}_{17}Cl$.

(i) Give the number of neutrons in the nucleus of $^{35}_{17}Cl$ (1mark)

(ii) Give that the two isotopes are present in ordinary chlorine in the ratio of three atoms of $^{35}_{17}Cl$ to one atom of $^{37}_{17}Cl$, Calculate the average atomic mass of chlorine. (4marks)

c. How would the chemical properties of the two isotopes compare? Give a reason for your answer (2marks)

d.

(i) What is radioactivity? (2marks)

(ii) When the nucleus of radium emits an alpha particle it decays to radon according to the equation:

$$^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$$

(1)Besides radon, what other particle is produced when radium decays? (1mark)

(2)How does the mass of the decaying atom compare with the masses of the Products? (1mark)

(3)Name the alpha particle in this equation? (1mark)

e. Explain how gamma rays are emitted. (4marks)

f. Give one use of gamma rays. (1mark)

g. Give one way of detecting radioactive particles. (1mark)